

NAME

io_cancel – cancel an outstanding asynchronous I/O operation

LIBRARY

Standard C library (*libc*, *-lc*)

Alternatively, Asynchronous I/O library (*libaio*, *-laio*); see VERSIONS.

SYNOPSIS

```
#include <linux/aio_abi.h> /* Definition of needed types */
#include <sys/syscall.h> /* Definition of SYS_* constants */
#include <unistd.h>

int syscall(SYS_io_cancel, aio_context_t ctx_id, struct iocb *iocb,
            struct io_event *result);
```

DESCRIPTION

Note: this page describes the raw Linux system call interface. The wrapper function provided by *libaio* uses a different type for the *ctx_id* argument. See VERSIONS.

The **io_cancel()** system call attempts to cancel an asynchronous I/O operation previously submitted with **io_submit(2)**. The *iocb* argument describes the operation to be canceled and the *ctx_id* argument is the AIO context to which the operation was submitted. If the operation is successfully canceled, the event will be copied into the memory pointed to by *result* without being placed into the completion queue.

RETURN VALUE

On success, **io_cancel()** returns 0. For the failure return, see VERSIONS.

ERRORS**EAGAIN**

The *iocb* specified was not canceled.

EFAULT

One of the data structures points to invalid data.

EINVAL

The AIO context specified by *ctx_id* is invalid.

ENOSYS

io_cancel() is not implemented on this architecture.

VERSIONS

You probably want to use the **io_cancel()** wrapper function provided by *libaio*.

Note that the *libaio* wrapper function uses a different type (*io_context_t*) for the *ctx_id* argument. Note also that the *libaio* wrapper does not follow the usual C library conventions for indicating errors: on error it returns a negated error number (the negative of one of the values listed in ERRORS). If the system call is invoked via **syscall(2)**, then the return value follows the usual conventions for indicating an error: -1, with *errno* set to a (positive) value that indicates the error.

STANDARDS

Linux.

HISTORY

Linux 2.5.

SEE ALSO

io_destroy(2), **io_getevents(2)**, **io_setup(2)**, **io_submit(2)**, **aio(7)**